

jcitest

Johansen cointegration test

Syntax

```
[h,pValue,stat,cValue,mles] = jcitest(Y)
[h,pValue,stat,cValue,mles] = jcitest(Y,Name,Value)
```

Description

Johansen tests assess the null hypothesis $H(r)$ of cointegration rank less than or equal to r among the `numDims`-dimensional time series in `Y` against alternatives $H(\text{numDims})$ (trace test) or $H(r+1)$ (maxeig test). The tests also produce maximum likelihood estimates of the parameters in a vector error-correction (VEC) model of the cointegrated series.

`[h,pValue,stat,cValue,mles] = jcitest(Y)` performs the Johansen cointegration test on a data matrix `Y`.

`[h,pValue,stat,cValue,mles] = jcitest(Y,Name,Value)` performs the Johansen cointegration test on a data matrix `Y` with additional options specified by one or more `Name,Value` pair arguments.

Input Arguments

<code>Y</code>	numObs-by-numDims matrix representing numObs observations of a numDims-dimensional time series y_t , with the last observation the most recent. <code>Y</code> cannot have more than 12 columns. Observations containing NaN values are removed. Initial values for lagged variables in VEC model estimation are taken from the beginning of the data.
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Name-Value Pair Arguments

Specify optional comma-separated pairs of `Name,Value` arguments. `Name` is the argument name and `Value` is the corresponding value. `Name` must appear inside single quotes (' '). You can specify several name and value pair arguments in any order as `Name1,Value1,...,NameN,ValueN`.

'model'	Character vector, such as 'H2', or cell vector of character vectors specifying the form of the deterministic components of the VEC(q) model of y_t: $\Delta y_t = C y_{t-1} + B_1 \Delta y_{t-1} + \dots + B_q \Delta y_{t-q} + DX + \varepsilon_t$ If $r < \text{numDims}$ is the cointegration rank, then $C = AB'$, where A is a numDims-by-r matrix of error-correction speeds and B is a numDims-by-r matrix of basis vectors for the space of cointegrating relations. X contains any exogenous terms representing deterministic trends in the data. For maximum likelihood estimation, it is assumed that $\varepsilon_t \sim \text{NID}(0, Q)$, where Q is the innovations covariance matrix. Values of model are those considered by Johansen [2]. <table border="1" data-bbox="418 401 1503 1060"> <thead> <tr> <th>Value</th><th>Form of $Cy_{t-1} + DX$</th></tr> </thead> <tbody> <tr> <td>'H2'</td><td>$AB'y_{t-1}$. There are no intercepts or trends in the cointegrating relations and there are no trends in the data. This model is only appropriate if all series have zero mean.</td></tr> <tr> <td>'H1*'</td><td>$A(B'y_{t-1} + c_0)$. There are intercepts in the cointegrating relations and there are no trends in the data. This model is appropriate for nontrending data with nonzero mean.</td></tr> <tr> <td>'H1'</td><td>$A(B'y_{t-1} + c_0) + c_1$. There are intercepts in the cointegrating relations and there are linear trends in the data. This is a model of <i>deterministic cointegration</i>, where the relations eliminate both stochastic and deterministic trends in the data. This is the default value.</td></tr> <tr> <td>'H*'</td><td>$A(B'y_{t-1} + c_0 + d_0 t) + c_1$. There are intercepts and linear trends in the cointegrating relations and there are linear trends in the data. This is a model of <i>stochastic cointegration</i>, where the relations eliminate stochastic but not deterministic trends in the data.</td></tr> <tr> <td>'H'</td><td>$A(B'y_{t-1} + c_0 + d_0 t) + c_1 + d_1 t$. There are intercepts and linear trends in the cointegrating relations and there are quadratic trends in the data. Unless quadratic trends are actually present in the data, this model might produce good in-sample fits but poor out-of-sample forecasts.</td></tr> </tbody> </table> Deterministic terms outside of the cointegrating relations, c_1 and d_1, are identified by projecting constant and linear regression coefficients, respectively, onto the orthogonal complement of A.	Value	Form of $Cy_{t-1} + DX$	'H2'	$AB'y_{t-1}$. There are no intercepts or trends in the cointegrating relations and there are no trends in the data. This model is only appropriate if all series have zero mean.	'H1*'	$A(B'y_{t-1} + c_0)$. There are intercepts in the cointegrating relations and there are no trends in the data. This model is appropriate for nontrending data with nonzero mean.	'H1'	$A(B'y_{t-1} + c_0) + c_1$. There are intercepts in the cointegrating relations and there are linear trends in the data. This is a model of <i>deterministic cointegration</i> , where the relations eliminate both stochastic and deterministic trends in the data. This is the default value.	'H*'	$A(B'y_{t-1} + c_0 + d_0 t) + c_1$. There are intercepts and linear trends in the cointegrating relations and there are linear trends in the data. This is a model of <i>stochastic cointegration</i> , where the relations eliminate stochastic but not deterministic trends in the data.	'H'	$A(B'y_{t-1} + c_0 + d_0 t) + c_1 + d_1 t$. There are intercepts and linear trends in the cointegrating relations and there are quadratic trends in the data. Unless quadratic trends are actually present in the data, this model might produce good in-sample fits but poor out-of-sample forecasts.
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'lags'	Scalar or vector of nonnegative integers indicating the number q of lagged differences in the VEC(q) model of y_t. Lagging and differencing a time series reduce the sample size. Absent any presample values, if y_t is defined for $t = 1:N$, then the lagged series y_{t-k} is defined for $t = k + 1:N$. Differencing reduces the time base to $k+2:N$. With q lagged differences, the common time base is $q+2:N$ and the effective sample size is $T = N - (q+1)$. Default: 0												
'test'	Character vector, such as 'trace', or cell vector of character vectors indicating the type of test to be performed. Values are 'trace' or 'maxeig'. The default value is 'trace'. Both tests assess the null hypothesis $H(r)$ of cointegration rank less than or equal to r. Statistics are computed using the effective sample size T and ordered estimates of the eigenvalues of $C = AB'$, $\lambda_1 > \dots > \lambda_d$, where $d = \text{numDims}$. - When the value is 'trace', the alternative hypothesis is $H(\text{numDims})$. Statistics are: $-T[\log(1 - \lambda_{r+1}) + \dots + \log(1 - \lambda_{\text{numDims}})]$ - When the value is 'maxeig', the alternative hypothesis is $H(r+1)$. Statistics are: $-T \log(1 - \lambda_{r+1})$												
'alpha'	Scalar or vector of nominal significance levels for the tests. Values must be between 0.001 and 0.999. Default: 0.05												

'display'	Character vector, such as 'off', or cell vector of character vectors indicating whether or not to display a summary of test results and parameter estimates in the Command Window.	
	Value	Display
	'off'	No display to the command window. This is the default if jcitest is called with only one output argument (h).
	'summary'	Display a summary of test results. Null ranks $r = 0:\text{numDims} - 1$ are displayed in the first column of each summary. Multiple tests are displayed in separate summaries. This is the default if jcitest is called with more than one output argument (that is, if pValue is computed), and is unavailable if jcitest is called with only one output argument (h).
	'params'	Display maximum likelihood estimates of the parameter values associated with the reduced-rank $\text{VEC}(q)$ model of y_t . This display is only available if jcitest is called with five output arguments (that is, if mles is computed). Displayed parameter values are returned in <code>mles.rn(m).paramVals</code> for null rank $r = n$ and test m .
	'full'	Display both summary and params.
Character vectors values are expanded to the length of any vector value (the number of tests). Vector values must have equal length.		

Output Arguments

h	numTests-by-numDims tabular array of Boolean decisions for the tests. Rows of h correspond to tests specified by the input arguments, and the software labels the rows $t_1, t_2, \dots, t_u$, where $u = \text{numTests}$. Variables of h correspond to different, maintained cointegration ranks $r = 0, \dots, \text{numDims} - 1$, and the software labels the variables $r_0, r_1, \dots, r_R$, where $R = \text{numDims} - 1$. To access results stored in h, for example, the result for test m of null rank n, use <code>h.rn(m)</code>. Values of h equal to 1 (true) indicate rejection of the null of cointegration rank r in favor of the alternative. Values of h equal to 0 (false) indicate a failure to reject the null.
pValue	numTests-by-numDims tabular array of right-tail probabilities of the test statistics. Rows of pValue correspond to tests specified by the input arguments, and the software labels the rows $t_1, t_2, \dots, t_u$, where $u = \text{numTests}$. Variables of pValue correspond to different, maintained cointegration ranks $r = 0, \dots, \text{numDims} - 1$, and the software labels the variables $r_0, r_1, \dots, r_R$, where $R = \text{numDims} - 1$. To access results stored in pValue, for example, the result for test m of null rank n, use <code>pValue.rn(m)</code>.
stat	numTests-by-numDims tabular array of test statistics, determined by the test name-value pair argument. Rows of stat correspond to tests specified by the input arguments, and the software labels the rows $t_1, t_2, \dots, t_u$, where $u = \text{numTests}$. Variables of stat correspond to different, maintained cointegration ranks $r = 0, \dots, \text{numDims} - 1$, and the software labels the variables $r_0, r_1, \dots, r_R$, where $R = \text{numDims} - 1$. To access results stored in stat, for example, the result for test m of null rank n, use <code>stat.rn(m)</code>.
cValue	numTests-by-numDims tabular array of critical values for right-tail probabilities, determined by the alpha name-value pair argument. jcitest loads tables of critical values from the file <code>Data_JCITest.mat</code>, then linearly interpolates test-critical values from the tables. Tabulated values were computed using methods described in [3]. Rows of cValue correspond to tests specified by the input arguments, and the software labels the rows $t_1, t_2, \dots, t_u$, where $u = \text{numTests}$. Variables of cValue correspond to different, maintained cointegration ranks $r = 0, \dots, \text{numDims} - 1$, and the software labels the variables $r_0, r_1, \dots, r_R$, where $R = \text{numDims} - 1$. To access results stored in cValue, for example, the result for test m of null rank n, use <code>cValue.rn(m)</code>.

mles	numTests-by-numDims tabular array of structures of maximum likelihood estimates associated with the VEC(q) model of y_t . Each structure contains these fields.	
	Field	Description
	paramNames	Cell vector of parameter names, of the form: {A, B, B1,...,Bq, c0, d0, c1, d1} Elements depend on the values of <code>lags</code> and <code>model</code> .
	paramVals	Structure of parameter estimates with field names corresponding to the parameter names in paramNames.
	res	T -by-numDims matrix of residuals, where T is the effective sample size, obtained by fitting the VEC(q) model of y_t to the input data.
	EstCov	Estimated covariance Q of the innovations process ε_t .
	eigVal	Eigenvalue associated with $H(r)$.
	eigVec	Eigenvector associated with the eigenvalue in eigVal. Eigenvectors v are normalized so that $v'S_{11}v = 1$, where S_{11} is defined as in [2].
	rLL	Restricted loglikelihood of Y under the null.
	uLL	Unrestricted loglikelihood of Y under the alternative.
Rows of mles correspond to tests specified by the input arguments, and the software labels the rows $t_1, t_2, \dots, t_u$, where $u = \text{numTests}$. Variables of mles correspond to different, maintained cointegration ranks $r = 0, \dots, \text{numDims} - 1$, and the software labels the variables $r_0, r_1, \dots, r_R$, where $R = \text{numDims} - 1$. To access results stored in mles, for example, the result for test m of null rank n , use <code>mles.rn(m)</code> . You can further access the fields of the structure using dot notation, for example, enter <code>mles.rn(m).paramNames</code> for the parameter names.		

Examples

collapse all

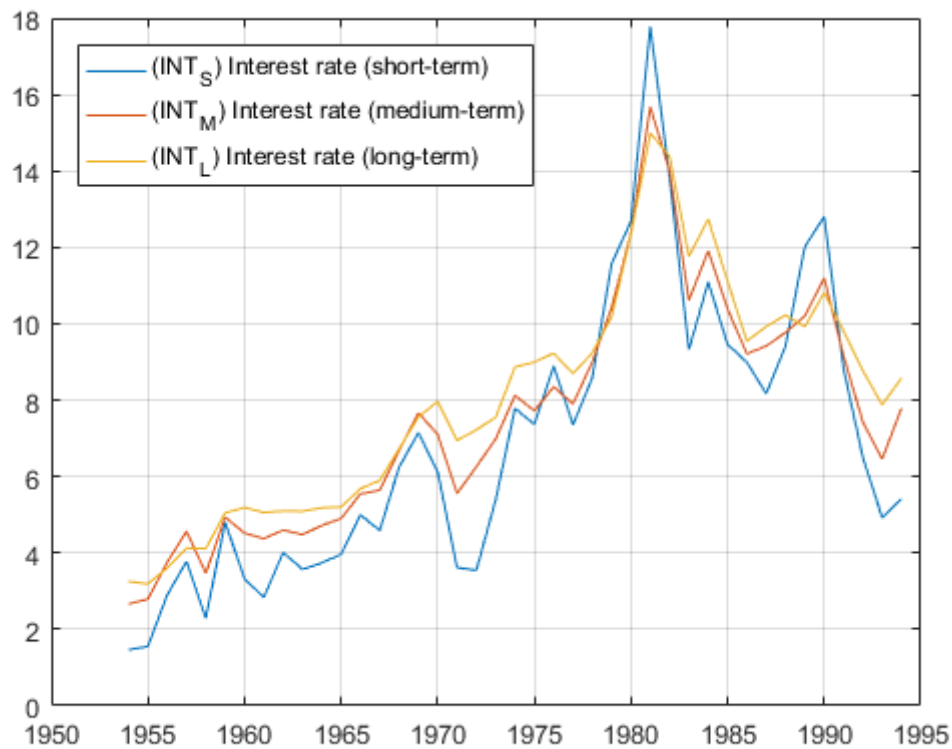
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Test Multiple Series for Cointegration Using jcitest

Load data on term structure of interest rates in Canada:

Open Script

```
load Data_Canada
Y = Data(:,3:end);
names = series(3:end);
plot(dates,Y)
legend(names,'location','NW')
grid on
```



Test for cointegration:

```
[h,pValue,stat,cValue,mles] = jcitest(Y,'model','H1');  
h,pValue
```

Results Summary (Test 1)

Data: Y

Effective sample size: 40

Model: H1

Lags: 0

Statistic: trace

Significance level: 0.05

r	h	stat	cValue	pValue	eigVal
0	1	37.6886	29.7976	0.0050	0.4101
1	1	16.5770	15.4948	0.0343	0.2842
2	0	3.2003	3.8415	0.0737	0.0769

h =

	r0	r1	r2
t1	true	true	false

pValue =

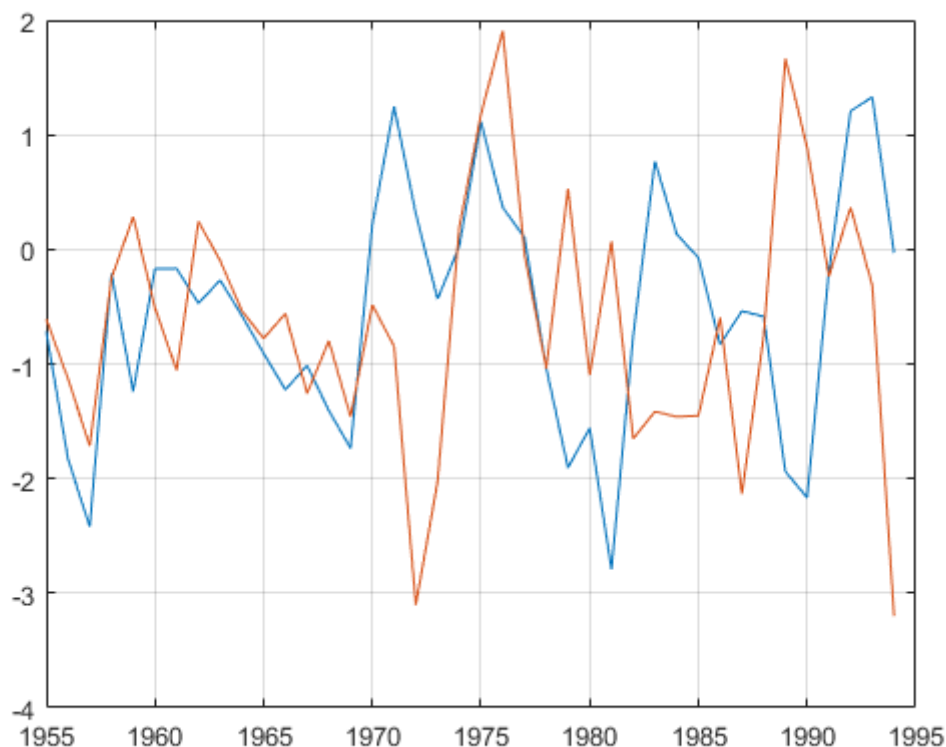
	r0	r1	r2
t1	0.0050497	0.034294	0.073661

Plot estimated cointegrating relations $B'y_{t-1} + c_0$:

```

YLag = Y(2:end,:);
T = size(YLag,1);
B = mles.r2.paramVals.B;
c0 = mles.r2.paramVals.c0;
plot(dates(2:end),YLag*B+repmat(c0',T,1))
grid on

```



More About

[collapse all](#)

▼ Algorithms

Time series in **Y** might be stationary in levels or first differences (i.e., $I(0)$ or $I(1)$). Rather than pretesting series for unit roots (using, e.g., [adftest](#), [pptest](#), [kpsstest](#), or [lmctest](#)), the Johansen procedure formulates the question within the model. An $I(0)$ series is associated with a standard unit vector in the space of cointegrating relations, and its presence can be tested using [jcontest](#).

If [jcitest](#) fails to reject the null of cointegration rank $r = 0$, the inference is that the error-correction coefficient C is zero, and the $VEC(q)$ model reduces to a standard $VAR(q)$ model in first differences. If [jcitest](#) rejects all cointegration ranks r less than `numDims`, the inference is that C has full rank, and y_t is stationary in levels.

The parameters A and B in the reduced-rank $VEC(q)$ model are not uniquely identified, though their product $C = AB'$ is. [jcitest](#) constructs $B = V(:,1:r)$ using the orthonormal eigenvectors V returned by [eig](#), then renormalizes so that $V' * S_{11} * V = I$, as in [2].

To test linear constraints on the error-correction speeds A and the space of cointegrating relations spanned by B , use [jcontest](#).

To convert $VEC(q)$ model parameters in the [mles](#) output to $VAR(q+1)$ model parameters, use [vec2var](#).

- [Cointegration and Error Correction Analysis](#)

References

[1] Hamilton, J. D. *Time Series Analysis*. Princeton, NJ: Princeton University Press, 1994.

[2] Johansen, S. *Likelihood-Based Inference in Cointegrated Vector Autoregressive Models*. Oxford: Oxford University Press, 1995.

[3] MacKinnon, J. G., A. A. Haug, and L. Michelis. "Numerical Distribution Functions of Likelihood Ratio Tests for Cointegration." *Journal of Applied Econometrics*. v. 14, 1999, pp. 563–577.

[4] Turner, P. M. "Testing for Cointegration Using the Johansen Approach: Are We Using the Correct Critical Values?" *Journal of Applied Econometrics*. v. 24, 2009, pp. 825–831.

See Also

[egcitest](#) | [jctest](#) | [vec2var](#)

Introduced in R2011a
